REVISION WORKSHEET

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MINISTRY OF EDUCATION

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وزارة التربية والتعليم

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Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 9 Advanced | Module 3 Lesson 2 | Motion with Constant Acceleration

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: $v_f = v_i + at$; $\Delta x = v_i t + \frac{1}{2}at^2$; $v_f^2 = v_i^2 + 2a\Delta x$; $\Delta x = \frac{1}{2}(v_i + v_f)t$. Use only for constant acceleration.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 9 book, printed pages 63–70.

P01. For constant acceleration, the velocity–time graph is:

A. Always horizontal B. Always a parabola
C. A straight line D. A circle

Answer

C. A straight line.

P02. Which relation does not require elapsed time explicitly?

A. $v_f = v_i + at$ B. $\Delta x = v_i t + \frac{1}{2}at^2$
C. $\Delta x = \frac{1}{2}(v_i + v_f)t$ D. $v_f^2 = v_i^2 + 2a\Delta x$

Answer

D. $v_f^2 = v_i^2 + 2a\Delta x$.

P03. A curved position–time graph for constant nonzero acceleration is:

A. A parabola B. Always a straight line
C. Always horizontal D. A vertical line

Answer

A. A parabola.

P04. For constant acceleration, average velocity over an interval equals:

A. The sum of the two speeds always B. Half the sum of initial and final velocities
C. Final velocity alone D. Acceleration multiplied by distance

Answer

B. Half the sum of initial and final velocities.

P05. Before applying the standard motion equations, you should check that:

A. Mass is zero B. Displacement is positive
C. Acceleration is constant over the interval D. Initial velocity is zero

Answer

C. Acceleration is constant over the interval.

P06. An object has initial velocity +4 m/s and constant acceleration +3 m/s² for 3 s. Find its final velocity.

A. 13 m/s B. 26 m/s
C. 6.5 m/s D. -13 m/s

Answer

A. 13 m/s. $v_f = v_i + at = 4 + 3 \times 3 = 13$ m/s.

P07. An object starts at +4 m/s and accelerates at +2 m/s² for 4 s. Calculate its displacement.

A. 64 m B. 32 m
C. 16 m D. -32 m

Answer

B. 32 m. $\Delta x = v_i t + \frac{1}{2}at^2 = 4 \times 4 + \frac{1}{2} \times 2 \times 4^2 = 32$ m.

P08. An object moves with initial speed 4 m/s and constant forward acceleration 2 m/s² through 6 m. Calculate its final speed.

A. 12.65 m/s B. 3.162 m/s
C. 6.325 m/s D. -6.325 m/s

Answer

C. 6.325 m/s. $v_f^2 = v_i^2 + 2a\Delta x = 4^2 + 2 \times 2 \times 6 = 40$; $v_f = 6.325$ m/s.

P09. A vehicle travels at 8 m/s and brakes at a constant acceleration −2.0 m/s². Find its stopping distance.

A. 32 m B. 8 m
C. -16 m D. 16 m

Answer

D. 16 m. $0 = v_i^2 + 2a\Delta x$; $\Delta x = -8^2/(2 \times (-2.0)) = 16$ m.

P10. A car slows uniformly from +12 m/s to rest in 4 s. Calculate its acceleration.

A. -3 m/s² B. -6 m/s²
C. -1.5 m/s² D. 3 m/s²

Answer

A. -3 m/s². $a = (0 - 12)/4 = -3$ m/s², opposite the initial velocity.

P11. State the meaning of each symbol in $v_f = v_i + at$.

Answer

v_f is final velocity, v_i initial velocity, a constant acceleration and t elapsed time, using consistent signs and units.

P12. Why is a sign convention important in one-dimensional motion?

Answer

It distinguishes opposite directions so velocities, accelerations and displacements combine consistently.

P13. Which equation is useful when final velocity is unknown but displacement and time are needed?

Answer

$\Delta x = v_i t + \frac{1}{2}at^2$, for constant acceleration.

P14. Why may total distance differ from displacement when using motion equations?

Answer

The equations give signed displacement. If the object reverses direction, total distance must add the lengths before and after reversal.

P15. What is the velocity–time slope for uniform acceleration?

Answer

It is constant and equals the acceleration.

P16. An object has initial velocity +6 m/s and constant acceleration +3 m/s² for 4 s. Find its final velocity.

Answer

$v_f = v_i + at = 6 + 3 \times 4 = 18$ m/s.

P17. An object has initial velocity +8 m/s and constant acceleration +3 m/s² for 5 s. Find its final velocity.

Answer

$v_f = v_i + at = 8 + 3 \times 5 = 23$ m/s.

P18. An object starts at +6 m/s and accelerates at +2 m/s² for 5 s. Calculate its displacement.

Answer

$\Delta x = vit + \frac{1}{2}at^2 = 6 \times 5 + \frac{1}{2} \times 2 \times 5^2 = 55$ m.

P19. An object starts at +8 m/s and accelerates at +2 m/s² for 6 s. Calculate its displacement.

Answer

$\Delta x = vit + \frac{1}{2}at^2 = 8 \times 6 + \frac{1}{2} \times 2 \times 6^2 = 84$ m.

P20. An object moves with initial speed 6 m/s and constant forward acceleration 2 m/s² through 9 m. Calculate its final speed.

Answer

$v_f^2 = v_i^2 + 2a\Delta x = 6^2 + 2 \times 2 \times 9 = 72$; $v_f = 8.485$ m/s.

P21. An object moves with initial speed 8 m/s and constant forward acceleration 2 m/s² through 12 m. Calculate its final speed.

Answer

$v_f^2 = v_i^2 + 2a\Delta x = 8^2 + 2 \times 2 \times 12 = 112$; $v_f = 10.58$ m/s.

P22. A vehicle travels at 12 m/s and brakes at a constant acceleration −2.0 m/s². Find its stopping distance.

Answer

$0 = v_i^2 + 2a\Delta x$; $\Delta x = -12^2/(2 \times (-2.0)) = 36$ m.

P23. A vehicle travels at 16 m/s and brakes at a constant acceleration −2.0 m/s². Find its stopping distance.

Answer

$0 = v_i^2 + 2a\Delta x$; $\Delta x = -16^2/(2 \times (-2.0)) = 64$ m.

P24. A car slows uniformly from +18 m/s to rest in 6 s. Calculate its acceleration.

Answer

$a = (0 - 18)/6 = -3$ m/s², opposite the initial velocity.

P25. A car slows uniformly from +24 m/s to rest in 8 s. Calculate its acceleration.

Answer

$a = (0 - 24)/8 = -3\ m/s^2$, opposite the initial velocity.

P26. A car starts from rest with $a = 2.0\ m/s^2$ for 5.0 s. Find final velocity and displacement.

Answer

$v_f = 0 + 2.0 \times 5.0 = 10\ m/s$. $\Delta x = \frac{1}{2} \times 2.0 \times 5.0^2 = 25\ m$.

P27. A ball moves at +12 m/s with acceleration $-3.0\ m/s^2$. Find when it first stops and its displacement by then.

Answer

From $0 = 12 - 3t$, $t = 4.0$ s. $\Delta x = 12 \times 4 + \frac{1}{2}(-3)(4^2) = 24\ m$.

P28. Why can the formula average velocity = (vi + vf)/2 fail for nonuniform acceleration?

Answer

It assumes a straight velocity–time graph, whose mean height is the endpoint average. A curved v–t graph can have a different mean height.

P29. Design a graph test for whether a cart has constant acceleration.

Answer

Measure velocity at equal time intervals and plot v against t. A straight line within uncertainty supports constant acceleration; its slope gives a.

P30. A calculated stopping distance is negative when the car was moving positively and braking. What should be checked?

Answer

Check signs in $v_f^2 = v_i^2 + 2a\Delta x$ and the algebra. For braking before reversal, a is negative but the displacement along the original positive direction is positive.